

Summary

Insights & Summary

After analysing the Housing Prices Dataset and building two regression models:

1. Which features influence house price the most? Based on the Random Forest feature importance scores, area (size of the property) is the single strongest predictor of house price. Other highly influential features include the number of bathrooms, number of stories, and whether the property is on the main road or in a preferred area. Amenities like air conditioning and a guestroom also positively impact price, but to a lesser degree.

2. How accurate was your model (in plain terms)? The Random Forest Regressor outperformed Linear Regression with a higher R^2 score, meaning it could explain a greater percentage of the variation in house prices. In simple terms, given a house's features, the Random Forest model can predict its price fairly closely. The Linear Regression model was simpler but slightly less accurate.

3. What surprised you in the data? The most surprising finding was that the number of bedrooms had less impact on price than expected. Buyers typically assume more bedrooms = higher price, but in this dataset, bathrooms and stories had a stronger correlation with price than bedrooms. This suggests that buyers value functional space and multi-floor living over raw bedroom count.

4. Business Recommendation Real estate businesses should focus marketing on area size, bathroom count, and location (main road / preferred area) as these are the top price drivers. For sellers looking to increase property value, investing in an additional bathroom or adding air conditioning offers a better return than adding bedrooms. The Random Forest model built here could be deployed as a simple price estimation tool on a real estate website to help buyers get instant, data-driven price estimates.